

Mathematical Modelling of Natural Systems

Assignment 2: Spatial Pattern Formation in Biology

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1 Our Non-Spatial system

We want to study how $u(x, t)$ and $v(x, t)$ work non-spatially in the system

$$\frac{\partial u}{\partial t} = a - u + u^2v + \frac{\partial^2 u}{\partial^2 x}, \quad (1)$$

$$\frac{\partial v}{\partial t} = \gamma b - \gamma u^2v + d \frac{\partial^2 v}{\partial^2 x}. \quad (2)$$

To do this we create variables $U = U(t)$, $V = V(t)$ such that

$$\frac{dU}{dt} = a - U + U^2V, \quad (3)$$

$$\frac{dV}{dt} = \gamma b - \gamma U^2V. \quad (4)$$

This is the non-spatial version of (1-2), where we just disregard the spatial derivatives as U and V don't depend on x .

Let $f(U, V) = a - U + U^2V$ and $g(U, V) = \gamma b - \gamma U^2V$. We shall use these later.

2 Co-existence

We will show that there is a unique coexistent equilibrium point of this system. Equilibrium points occur when $\frac{dU}{dt} = \frac{dV}{dt} = 0$. Hence, we have:

$$a - U + U^2V = 0 \quad (5)$$

$$\gamma b - \gamma U^2V = 0 \quad (6)$$

Now, (6)+ γ (5) gives:

$$\gamma b - \gamma U^2V + \gamma(a - U + U^2V) = 0$$

$$\implies \gamma b + \gamma(a - U) = 0.$$

Dividing by γ , which we can do as $\gamma > 0 \implies \gamma \neq 0$ gives:

$$b + a - U = 0$$

$$\implies U = b + a.$$

Substituting this back into (5), we obtain:

$$a - (b + a) + (b + a)^2V = 0$$

$$\implies (b + a)^2V = b$$

$$\implies V = \frac{b}{(b+a)^2}.$$

Now, a and b are fixed parameters, and so these stationary points of the system (3-4) are unique. Let this fixed point be $(U(t), V(t)) = (u_*, v_*) = (b + a, \frac{b}{(b+a)^2})$.

3 Stability

The stability of an equilibrium point (for this non-spatial system) is determined by the trace of the Jacobian of the system. More precisely, an equilibrium point is stable if and only if $Det(J) > 0$ and $Tr(J) < 0$, where J is the Jacobian evaluated at (u_*, v_*) [1].

We calculate the Jacobian: $J(U, V) := \begin{pmatrix} f_U & f_V \\ g_U & g_V \end{pmatrix}$,

where $f_U(U, V) = -1 + 2UV$, $f_V(U, V) = U^2$, $g_U(U, V) = -2\gamma UV$, $g_V(U, V) = -\gamma U^2$, are partial derivatives of f and g in U and V .

Hence, we have:

$$J(U, V) := \begin{pmatrix} -1 + 2UV & U^2 \\ -2\gamma UV & -\gamma U^2 \end{pmatrix}.$$

Now, $\text{Det}(J(U, V)) = (-1 + 2UV)(-\gamma U^2) - U^2(-2\gamma UV) = \gamma U^2 > 0$ for all (U, V) , so we do not obtain any constraints from this. Also, $\text{Tr}(J(U, V)) = (-1 + 2UV) + (-\gamma U^2)$.

So, (u_*, v_*) is stable

$$\iff \text{Det}(J((u_*, v_*))) > 0 \text{ and } \text{Tr}(J((u_*, v_*))) < 0$$

$$\iff -1 + 2u_*v_* - \gamma u_*^2 < 0$$

$$\iff 1 - 2u_*v_* + \gamma u_*^2 > 0$$

$$\iff 1 - 2(a+b)\left(\frac{b}{(a+b)^2}\right) + \gamma(a+b)^2 > 0$$

$$\iff 1 - \frac{2b}{a+b} + \gamma(a+b)^2 > 0$$

$$\iff (a+b) - 2b + \gamma(a+b)^3 > 0$$

$$\iff \gamma(a+b)^3 > b-a, \text{ as required.}$$

4 My Parameter Values

I have been given the parameters $\gamma_p = 1$, $a_p = 0.1$, $b_p = 0.9$, $d_p = 12$. From these parameters, $\gamma(a+b)^3 = 1(0.1+0.9)^3 = 1$, whereas $b-a = 0.9-0.1 = 0.8$. So indeed, we have $\gamma(a+b)^3 > b-a$. Hence, the point $(u_*, v_*) = (b+a, \frac{b}{(b+a)^2}) = (1, 0.9)$ is stable in the non-spatial system by part 3.

5 Adding Space

Suppose there are solutions of (1-2) of the form

$$u(x, t) = u_* + \hat{u} = u_* + u_0 \exp(i\kappa x + \sigma t) \quad (7)$$

$$v(x, t) = v_* + \hat{v} = v_* + v_0 \exp(i\kappa x + \sigma t) \quad (8)$$

We calculate the partial derivatives:

$$\frac{\partial u}{\partial t} = \sigma \hat{u}, \quad \frac{\partial v}{\partial t} = \sigma \hat{v}, \quad \frac{\partial^2 u}{\partial x^2} = -\kappa^2 \hat{u}, \quad \frac{\partial^2 v}{\partial x^2} = -\kappa^2 \hat{v}.$$

Plugging these into (1-2), multiplying these out and neglecting terms which include $\hat{u}^2, \hat{v}^2, \hat{u}\hat{v}$, as these terms are negligible for small t (as perturbations as u_0 and v_0 are very small initial perturbations, in comparison to much larger linear terms), we get:

$$\begin{aligned} (1) &\implies \sigma \hat{u} = a - (u_* + \hat{u}) + (u_* + \hat{u})^2(v_* + \hat{v}) - \kappa^2 \hat{u} = a - (u_* + \hat{u}) + (u_*^2 + 2u_*\hat{u} + \hat{u}^2)(v_* + \hat{v}) - \kappa^2 \hat{u} \\ &\approx a - (u_* + \hat{u}) + (u_*^2 + 2u_*\hat{u})(v_* + \hat{v}) - \kappa^2 \hat{u} \\ &= a - (u_* + \hat{u}) + u_*^2 v_* + u_*^2 \hat{v} + 2u_* v_* \hat{u} + 2u_* \hat{u} \hat{v} - \kappa^2 \hat{u} \\ &\approx a - u_* - \hat{u} + u_*^2 v_* + u_*^2 \hat{v} + 2u_* v_* \hat{u} - \kappa^2 \hat{u}. \end{aligned}$$

Now as (u_*, v_*) is an equilibrium point of (3-4), $a - u_* + u_*^2 v_* = 0$, (by equation (5)) so

$$\sigma \hat{u} \approx u_*^2 \hat{v} + 2u_* v_* \hat{u} - \hat{u} - \kappa^2 \hat{u} = (2u_* v_* - 1 - \kappa^2, u_*^2) \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix}.$$

$$\text{Similarly, (2)} \implies \sigma \hat{v} \approx (-2\gamma u_* v_*, -\gamma u_*^2 - \kappa^2 d) \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix}.$$

We define a new system as

$$\sigma \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} = \begin{pmatrix} -1 + 2u_* v_* - \kappa^2 & u_*^2 \\ -2\gamma u_* v_* & -\gamma u_*^2 - d\kappa^2 \end{pmatrix} \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} = M \begin{pmatrix} \hat{u} \\ \hat{v} \end{pmatrix} \quad (9)$$

which is roughly equivalent to the old system, and is a better approximation as $t \rightarrow 0$.

6 Dispersion Relation

We plot the dispersion relation for our system by calculating the eigenvalues of the matrix M in (9) for different values of κ . What we plot is the largest real part of the 2 eigenvalues that we get. When this is greater than 0 it indicates that patterns of this wavelength could form given an initial perturbation, as it means that these parts of the wavelength decomposition of an initial perturbation grow, whilst all other parts shrink to 0, leaving these wavelengths dominating.

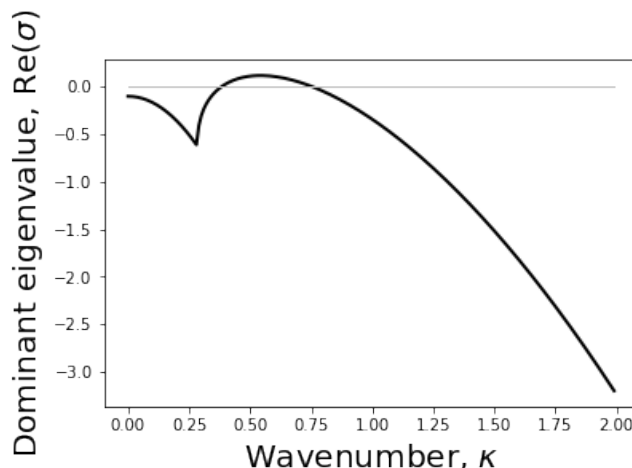


Figure 1: Dispersion relation for the given parameter values.

7 Animation of Numerics

See “Animation 1”, which plots space (x) along the horizontal axis against the changing densities of $u(x, t)$, $v(x, t)$ along the vertical axis. This changes in time. From this, we can approximate the wavelength λ of the pattern forms by counting the number of waves over some space (i.e there are roughly 8.5 waves in 100 units of space, so wavenumber $\lambda \approx \frac{100}{8.5}$). The wavenumber κ is then $\kappa = \frac{2\pi}{\lambda}$. In this case we work out $\kappa \approx 0.534$. If we look at the dispersion relation in figure 1, this falls into the positive values of κ , as our analysis has suggested it would. In fact, we can make this method more precise by using the method outlined in Murray’s *Mathematical Biology II*[2] (pages 85-86, and we can use this as the systems spoken about are very similar, and identical when we let $\gamma = 1$), we find the range of values of possible wavenumbers κ of this system to be $[0.382, 0.755]$, and so our estimated wavenumber falls into this range. This also is what it looks like the range is according to our plot. We can also find the critical wavenumber as 0.5203 which is very close to our observed roughly calculated wavenumber.

We note here that “Animation 2” and “Animation 3” have the same axes as “Animation 1”, and are used in the next section. Between these animations, only the value of d changes:

Animation 1: $d = 12$ Animation 2: $d = 20$ Animation 3: $d = 9$

8 Investigation

The regularity in waves appearing in “Animation 1” show us that patterns will be evenly distributed along the section of fur we are looking at. The repeated amplitude of the waves indicate that patterns will all be of the same colour. The bounds on the wavenumber in the dispersion relation means we do not expect to see patterns which are too close together nor too far apart. Indeed, patterned animals

(zebras, tigers) tend to have fur with evenly distributed patterns of the same colour that are not too close or far apart.

The wave stabilising after some time indicates that the patterns will not change thereafter. From this we can deduce that the patterns will take some time to develop, and then remain constant for the rest of time. The proverb “a leopard never changes it’s spots”, can therefore be seen as true, if this system were to model a leopard’s coats.

By changing parameters individually, and using the condition for stability in the non-spatial system given in Q3, we can use Newton Raphson to calculate lower bounds for a, b and γ : $a > 0.048$, $b > 0.779$ and $\gamma > 0.8$. From this we learn that these parameters cannot be too small if we want patterns to form. These parameters are associated with the flux of chemicals to the surface, and it makes sense that small amounts of chemicals coming to the surface would mean that no patterns emerge.

By using equations (66-69) in the appendix of the notes, which gives criteria for patterns to form in the spatial system, we calculate more bounds on the spatial system: $0.048 < a < 0.167$, $0.779 < b < 1.131$, $0.8 < \gamma < 1.4$, $d > 8.57$. In fig.2 we see the dispersion relation only just kisses a point at 0 for these new bounds, supporting our calculations. The upper bounds on a and b mean that the density of one chemical cannot be too much larger than the other if we want patterns to form- there needs to be some balance between them in order for patterns to form.

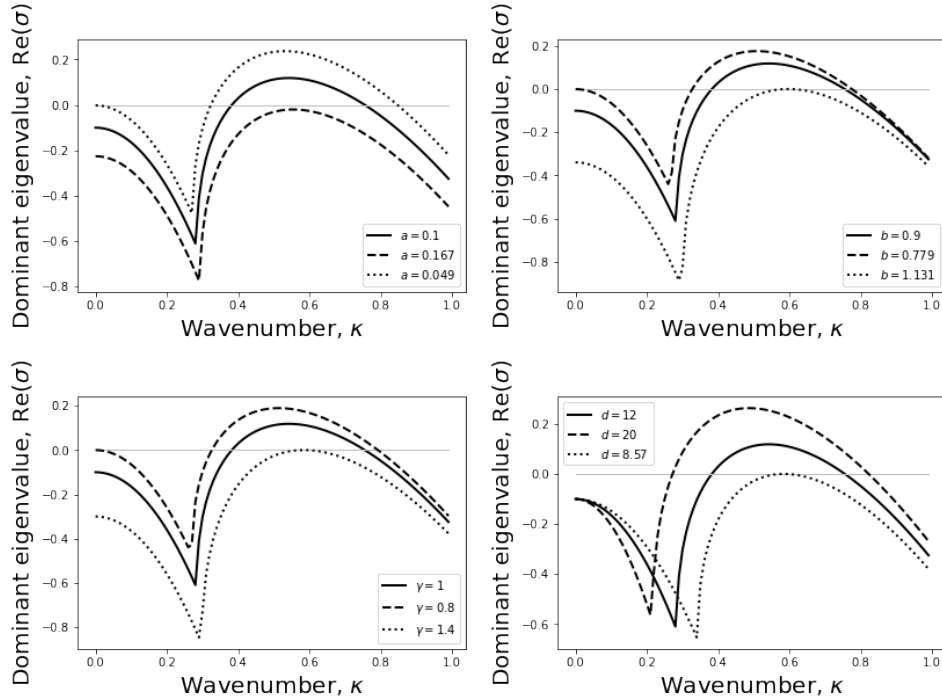


Figure 2: This shows the range of dispersion relations which create patterns by changing individual parameters. The solid black line is unchanged from fig.1. The dotted lines show the bounds for patterns to form. a) changes a , b) changes b , c) changes γ , d) changes d . For this parameter, there is no upper bound, so I have used the value of d used in ”Animation 3”.

In *Mathematical Biology II*[2], a Schnakenberg diffusion-reaction system is defined very similarly to our system. In this system, γ “is proportional to the linear size of the spatial domain”. If we relate γ to the size of the animal, the bounds on γ suggest that both small and large animals should be expected to be uniform in colour. Indeed, this is what we tend to see (mice, elephant). Murray also relates γ to the size of a developing limb bud in mammalian embryos. Perhaps we could suggest that this is one of the causes of Panda bears having stripes in contrast to the uniformly coloured sloth

bears, despite having roughly the same size[3]. Indeed, “Pandas have the shortest leg-to-body ratio among the world’s eight living bear species”[4]¹. Hence, we could suggest that the size of the Panda’s legs cause γ to be within the range for patterns to form, whereas the slightly bigger limbs of the sloth bear cause it’s γ to fall outside that range.

As d is the parameter which affects the rate of chemical diffusion of v , we can interpret the lower bound as being caused by the rate of chemical diffusion of v being too slow. Intuitively, it makes sense that a slow diffusion of v will neither activate, nor inhibit u enough to cause patterns. We compare animations 1, 2 and 3 which changes d between 12, 20 and 9 respectively. “Animation 2” has a higher value of d , and the patterns form more quickly than in animation 1 with greater differences between densities of u and v at peaks and troughs. Similarly, in “Animation 3” with lower d we see very small differences in u and v that take a long time to develop. A greater distance between densities perhaps indicate more defined patterns. We could suggest that a variation in this parameter causes the difference in the colouration of snow leopards, which have spots that tend to blend into the background of their grey fur in comparison to the African leopard, which has very defined spots, despite both species being of similar sizes and shapes and type of pattern (see fig.3). This would make sense, as the colder environment of the snow leopard would slow the rate of diffusion (as the particles diffusing have less energy) and so it would make sense if d was smaller for the snow leopard. If this were the case, from our observations in the animations, we would expect that patterns on the embryo of a snow leopard take longer to form than on the embryo of an African Leopard.



Figure 3: From [6], showing a comparison of patterns on a Snow Leopard (left) and an African Leopard (right). The spots on the Snow Leopard are less defined than the spots on the African Leopard

References

- [1] Johnathon Potts *MAS222: Differential Equations* lecture notes, University of Sheffield, 2019-2020.
- [2] J.D. Murray *Mathematical Biology II* Springer-Verlag New York
- [3] Comparing the size of the Sloth bear and the giant panda bear on Dimensions.com:
<https://www.dimensions.com/element/sloth-bear>
<https://www.dimensions.com/element/giant-panda-bear>
- [4] Susan Milius, on panda’s having the smallest legs of any variety of bear.
<https://www.sciencenewsforstudents.org/article/pandas-use-their-heads-as-a-kind-of-extra-limb-for-climbing>
- [5] J.D. Murray, *A Pre-pattern formation mechanism for animal coat markings*, Journal of Theoretical Biology, Volume 88, Issue 1, 1981, Pages 161-199, ISSN 0022-5193, [https://doi.org/10.1016/0022-5193\(81\)90334-9](https://doi.org/10.1016/0022-5193(81)90334-9). (<https://www.sciencedirect.com/science/article/pii/0022519381903349>)
- [6] Images of leopards found from <https://lprd.design/blogs/news/cheetah-or-leopard-print>

¹I could not actually find any research on this matter. To understand this properly, we would need to get data about the embryonic shape of the two bears. This, in general, is not well studied[5].